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A hybrid CNN-GRU based probabilistic model for load forecasting from individual household to commercial building

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Abstract

In load forecasting, build load is relatively difficult to predict due to its variability and uncertainty. This study intends to develop a hybrid deep learning models and quantile regression loss function for building load prediction. Then, through the numerical study for the individual and commercial building, it is provided to demonstrate the practicality of proposed method which showed the accuracy with lower loss than existing models. In academic contribution, this study proves that Convolutional Neural Network (CNN) can extract useful information from high uncertainty power load, and Gated Recurrent Unit (GRU) has the benefit in the times-series forecasting. Furthermore, the proposed hybrid framework outperforms the conventional LSTM and GRU neural network. In practical contribution, the issues of reducing energy waste and demand side management can be improved through more accurate load forecasting. Also, RE100 company can realize the risk of the power purchasing planning and customized decision-making through prediction intervals.
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Keywords: Deep learning; Hybrid model; Energy forecasting; Building load; Prediction interval

1. Introduction

The recent rapid economic and population growth is accelerating the growth of electricity consumption. According to the Global Energy Outlook 2022 (GEO 2022), the average worldwide electricity will rise more than 50% by 2050 [1]. Electrification of buildings becomes one of the major measures to achieve net zero carbon feature, so there has been a growing interest in building load forecasting issues, especially in the intelligent management.

With the rise of building's awareness of smart grid-related issues, and deep learning in recent years, scholars have begun to apply deep learning techniques to smart grids, cities, and homes applications to respond to the advent of smart era. In order to support smart grid applications in the future, effective load forecasting techniques for power

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users are receiving increasing attention. Individual households and commercial buildings have totally different load patterns. The building loads show greater volatility especially the short-term due to the policies, different living habits and numbers which leads to higher uncertainty for traditional models to deal with. Data with time series information which is a complex nonlinear system also makes forecasting harder. Traditional point load forecasting can only provide the expected values of future loads with limited ability to interpret the uncertainty of the prediction results, which leads to inaccurate prediction, energy wasted and so on. Hence, the purpose of this study is to propose a hybrid deep learning-based model integrated in both the individual and commercial buildings, which not only can make more accurate predictions compared to traditional methods, but also inform the power consumers of the uncertainty of the prediction to help them allocate the electricity and planning the purchasing.

With the explosive growth of accumulated power data from smart grid and the development of machine learning methods, there are many researches focuses on the establishment of various models to different building types in various aspects. Authors of [2] had been applied Artificial Neural Network (ANN) on short-term load forecasting. However, complicated and high variety data may need more powerful time series model to make accurate predict. Feature extraction as well as energy consumption patterns of building energy prediction is very important when applying the deep learning methods. Based on the challenges of feature extraction and time series data, therefore, we proposed a hybrid deep learning- based model with Convolutional Neural Network (CNN) for feature extraction and Gated Recurrent Unit (GRU) for prediction integrated in the scenario of short-term building load, which can make more accurate predictions compared to traditional methods, and assist power company and consumer as an important foundation for energy management. Moreover, in order to overcome the limitations of data-driven techniques to fit the different operating conditions of the building, a prediction interval will be provided.

2. Literature review

2.1. Deep learning in time series load forecasting

We have compiled several scholars who applied deep learning to the field of short-term electric loads. Authors of [3] presented a load forecasting for multiple households with different consumption patterns using Bayesian networks. Authors of [2] applied ANN to predict the future electric load through a factorized conditionally restricted Boltzmann machine (FCRBM), and put forward the proposed genetic wind drive (GWDO) optimization algorithm for short-term load forecasting. However, considering the variability and high uncertainty, building load need a model which not only has the ability to forecast energy consumption accurately but also has the ability to tell uncertainty.

Basically, traditional ANN, which have been known as Multilayer Perceptron (MLP) already have a good performance in forecasting. However, the ANN architecture alone may fail to learn all the features effectively. Thus, CNN was proposed with the concept of weight sharing. Although CNN is more commonly used in image recognition or classification, it has been proved to be capable of extracting the features and invariant structures exhibited in time series data. However, to make CNN learn non-linear dependencies, large number of layers and filters are needed, and this may lead to overfitting. Authors of [4] proposed a hybrid modeling technology based on the combination of DL, CNN and SVM, to carry out short-term load forecasting challenges, and can effectively deal with the nonlinear complexity and short-term dependence of power consumption time series data. It is proved that CNN truly can be applied in time series forecasting but still have some limits and drawbacks.

Recurrent neural network (RNN) is widely used in time series forecasting due to its consideration of the time sequence. When RNN is implemented, it is often used in conjunction with Long Short-Term Memory (LSTM). Unlike traditional neural networks, LSTMs have memory blocks as neurons [5]. In each block, there are several gates which control the state of the block and the output. Those gates are input gate, forget gate and output gate. LSTMs are trained by backpropagation through time, it is similar to backpropagation, but it considers the condition of time in turn.

Several scholars choose LSTM for prediction. Authors of [6] used the wavelet sequence model through the LSTM network to classify electrocardiogram (ECG) signals. Authors of [7] made a significant progress in identifying human behavior in videos. We can indeed see from the past literature that LSTM can provide a solution by merging memory cells that can update the previous hidden state to preserve long-term memory. Similar to LSTM, GRU is a recurrent neural network which has also been proposed to solve the problems of long-term memory and gradient disappearance in backpropagation [8]. Compared with LSTM, there is one less gate in GRU. There are fewer

parameters than LSTM, but it can also implement functions equivalent to LSTM. Authors of [9] did the load forecasting by preprocess load data through GRU to reduce the interference of electricity consumption characteristics and make predictions.

Above literature show that both LSTM and GRU can provide a solution by merging memory cells that can update previously hidden states to preserve long-term memory. However, if only LSTM or GRU is used alone as the forecasting model, we are likely to miss the message given by the feature in data. Therefore, recent practical applications have begun to adopt hybrid methods.

2.2. Hybrid deep learning-based models in time series forecasting

In recent years, there has been an increasing interest in combining different deep learning models to extract temporal and spatial features. Among these hybrid time series deep learning models, the combination of CNN and LSTM has shown remarkable performance in various applications, such as air quality forecasting [10] and energy forecasting [11]. However, due to the high volatility of household data, the performance of these models may be limited. To address this issue, researchers have proposed various hybrid deep learning models to improve the prediction accuracy of household electricity consumption. Authors of [12] combined CNN with LSTM and proposed a hybrid deep learning neural network framework to further improve the prediction accuracy of household electricity consumption prediction, and compared the results with four existing methods to show this ability. Authors of [13] combined RNN and 1-dimensional CNN as recurrent inception convolution neural network (RICNN) for multi short-term building load forecasting, and RICNN model outperforms the benchmarked multi-layer perception, RNN, and 1-D CNN. Authors of [14] utilized the WPD and LSTM deep learning model to predict the photovoltaic power generation per hour at five-minute intervals. Authors of [15] proposed a deep RNN model with GRU to forecast the load demand of residential buildings over a short to medium time period. Authors of [16] compares the performance of time series, machine learning, and hybrid models for accurate peak load-forecasting from the national level in Korea. The results indicate that the hybrid models, particularly the LSTM-based ones, outperform the traditional time series-based model, indicating that machine learning or hybrid models could improve Korea's peak load-forecasting performance. Building-level load forecasting is distinct from national level forecasting as it pertains to a particular location and usage pattern influenced by various factors such as occupancy, equipment, and building design. The Authors of [17] proposed a hybrid method using Prophet and LSTM models to overcome the limitations of existing short-term load forecasting models, which outperforms standalone models on the basis of reduced errors and least computation time, as verified by real-time electrical load data from 2014 to 2021. It can be found from the above literature that the hybrid model has been verified to be effective from many perspectives, especially in power load. Among them, there is still a lack of explanation for the uncertainty of the forecast results. Therefore, a hybrid deep learning model with powerful predictive ability and probability information has been developed to predict household load consumption in this study.

At present, in the field of time series prediction based on the CNN-GRU hybrid deep learning model, Authors of [18] established a 3D CNN-GRU model to predict household electricity consumption per hour, and [19] used the hybrid network technology of CNN-GRU-EWO to model the daily power generation. [18,19] are different from the prediction of household electricity consumption every five minutes in this study. Finally, Authors of [20] also uses the GRU-CNN hybrid neural network model for power prediction. The data set used in the study is the electricity consumption data of a certain city in China, which is different from the building level consumption in this study.

The difference between this research, [21,22] is that this research accurately predicts short-term household electricity consumption through a more complex CNN-GRU model. Authors of [21] used the concept of compound prediction. The proposed prediction model further provides new insights into the prediction interval estimation to reflect the potential uncertainties of power system planning and operation. Authors of [22] proposed a GRU-CNN hybrid neural network model which can make fuller use of data and achieve more accurate short-term load forecasting. Finally, compared with [23], although the CNN-GRU model has been used, it is used to predict the power consumption of buses, and the method of interval prediction is not used. Therefore, the method provided in this study still has reference value in the field of building time series electricity consumption forecasting.

2.3. Probabilistic forecasting

As the uncertainty of the data and environment increases, point estimation can no longer sufficient for forecasting, and probabilistic forecasting will become more and more important in the form of predictive probability distributions.

Scholars started to apply probabilistic forecasting instead of deterministic forecasting in different aspects, and related studies have begun to focus on issues including renewable energy generation, air pollution, electricity price, power consumption, and so on. Due to the randomness and intermittence involved, probabilistic forecasting is suitable for those issues. In latest research in probabilistic electricity price forecasting, Authors of [24] introduced a new approach to simulate electricity prices and provided probabilistic forecasts.

Some scholars have tried to combine probabilistic prediction with deep neural networks and apply them to their research. Authors of [25] used density estimation ANN to introduce probabilistic prediction, implemented two different probabilistic ANN models, and compared them on three different data sets. Authors of [26] uses LSTM to predict the probability of electricity. Its model can be used as a decision-making for power retailers to participate in the short-term power market. Authors of [27] proposed an autoregressive RNN model on a time series with probabilistic forecasting. From the above literature, only basic artificial neural networks are used, and the field of probabilistic forecasting has not yet used hybrid models which may lead to better results.

3. The proposed CNN-GRU hybrid model with prediction interval

Aiming to solving the high volatility and uncertainty of building load forecasting problem, we propose a hybrid model which combining both CNN and GRU in order to provide more accurate and robust forecasting result. Fig. 1 illustrates the structure of predicting power load by using proposed CNN-GRU architecture with prediction interval which will be applied to the household individual and commercial building respectively.

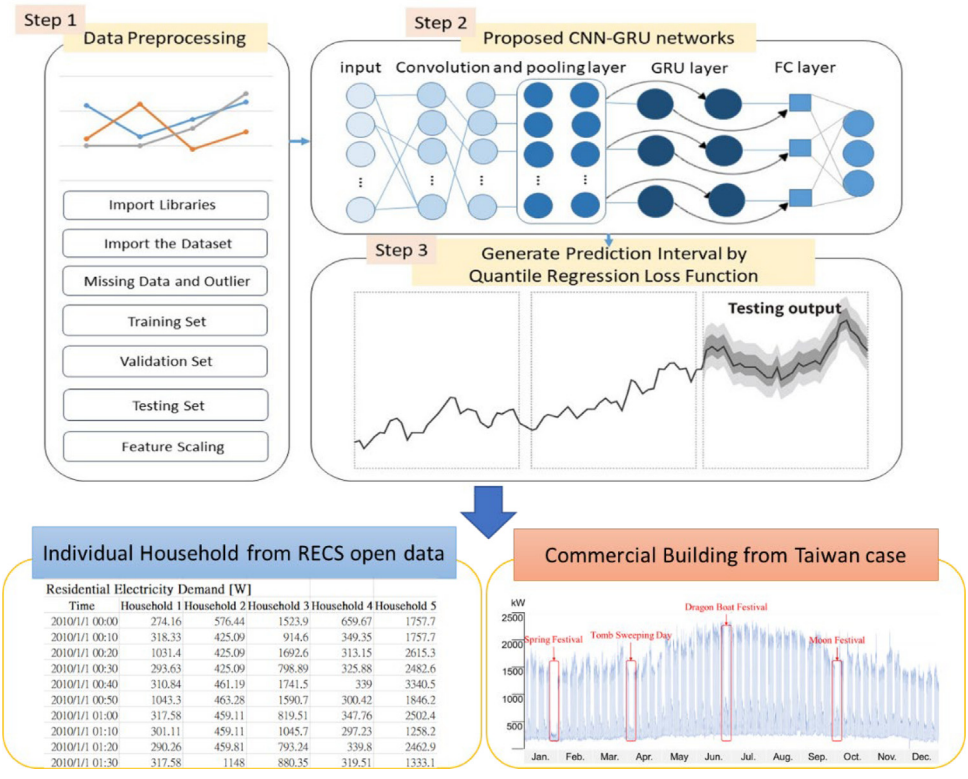


Fig. 1. The proposed building power consumption prediction structure.

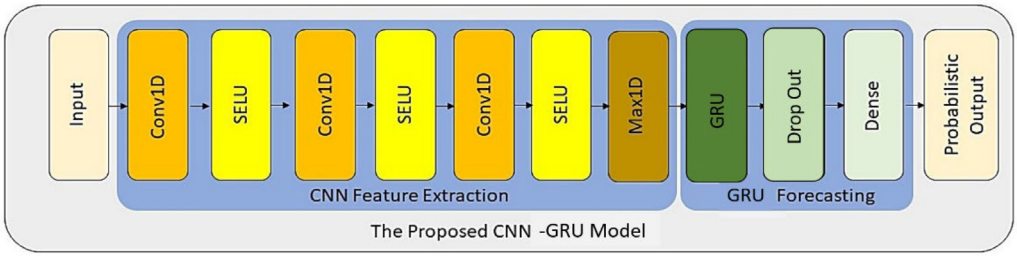


Fig. 2. The solution flowchart of proposed CNN-GRU hybrid Model.

The overall process is roughly as follows. First, the spatial features in our time series data will be extracted through convolution and pooling layers. Then, the temporal feature of the data will be modeled by the GRU. Afterwards, the results learned from the model will be used to predict household power consumption as an interval on the testing data.

The solution flowchart is shown in Fig. 2. The first half of the model is CNN, which is utilized for feature extraction, containing three Conv1D layers and one Max1D layer. And the remaining is GRU forecasting, which is utilized to analyze the features extracted by CNN and estimate the household power load of the next time period, containing one GRU and one Dropout. We will introduce the details in each part in the following subsection.

3.1. Convolutional neural network

CNN models pass input through a series of convolution layers with filters (or so called kernels), pooling layers, fully connected layers (FC) and apply an activation function to make classification. Commonly, convolution layers make neurons share weights and reduce the number of parameters; pooling layers ensure translation invariance of the learned features, which make CNN more powerful and easier to train than traditional neural networks.

In the first half of our proposed model, convolution layers are used to extract local data change trend features. This characteristic of CNN can help extract information from a single family without being affected by its uncertainty. The CNN part contains three one-dimensional (1D) convolution layers and one 1D max pooling layers. Scaled Exponential Linear Units (SELU), as shown in (1), is utilized as the activation function. There are two reasons to choose SELU as activation function instead of Rectified Linear Unit (ReLU), as shown in (2). First, SELU has a better convergence ability compared to other activation functions. Second, it can avoid the problem of gradient vanishing.

$$\text{SELU} = \lambda \begin{cases} x & if\ x > 0 \\ \alpha e^{x} - \alpha & o.w. \end{cases} \tag{1}$$

$$\text{ReLU} = \max(0, x) \tag{2}$$

3.2. Gate recurrent unit

The rest part contains only one GRU layer. It stores time information about important characteristics of power load extracted through CNN, and passing its output through the fully-connected layer and the sigmoid activation function, as shown in (3), to produce the final output. Compared with RNN, it prevents the problem of gradient vanishing and is able to store long term information, which makes it more suitable than other methods.

$$S(x) = \frac{1}{1 + e^{-x}} \tag{3}$$

GRU cells do not have separated cells like LSTM [8]. Memory cell consists of four parts: input gate, neuron with self-circulating connections, forget gate and an output gate. Notations are denoted as follows:

h_t: hidden layer vectors in GRU with $t = 1, 2, \ldots, t-1, t$

x_t: output vectors in GRU with $t = 1, 2, \ldots, t-1, t$

z_t: update gate in GRU with $t = 1, 2, \ldots, t-1, t$

r_t: reset gate in GRU with $t = 1, 2, \ldots, t-1, t$
W_z: the parameter matrices of update gate in GRU
W_r: the parameter matrices of reset gate in GRU
W_h: the parameter matrices of hidden layer in GRU
b_z: bias vector of update gate in GRU
b_r: bias vector of reset gate in GRU
b_h: bias vector of hidden layer in GRU
σ, tanh: activation functions
The computation can be denoted as follow

$$z_t = \sigma\left(W_z \cdot [h_{t-1}, x_t] + b_z\right) \tag{4}$$

$$\mathrm{r_t} = \sigma\left(\mathrm{W_r} \cdot [\mathrm{h_{t-1}}, \mathrm{x_t}] + \mathrm{b_r}\right) \tag{5}$$

$$\widetilde{h}_t = \tanh\left(W_h \cdot [r_t \odot h_{t-1}, x_t] + b_h\right) \tag{6}$$

$$h_t = (1 - z_t) \odot h_{t-1} + z_t \odot \widetilde{h}_t \tag{7}$$

3.3. Prediction interval

Previous part of our model tries to construct an accurate point estimator, and in the last step, our model will generate a 5% to 95% prediction interval based on the result of previous CNN-GRU by applying quantile regression loss function. A prediction interval is a quantification of the uncertainty on a prediction. It provides a probabilistic upper and lower bounds, and a quantile is the value below which a fraction of observations in a group falls. Given a prediction y_i^p and outcome y_i, the mean regression loss for a quantile q is shown as Eq. (8)

$$L\left(y_i^p, y_i\right) = \max\left[q\left(y_i^p - y_i\right), (q-1)\left(y_i^p - y_i\right)\right] \tag{8}$$

In above, as q has a value between 0 and 1, and the larger the value of q, the more over-predictions are penalized compared to under-predictions. The reason why choosing quantile loss is that prediction interval from least square regression assumes that residuals (true-predicted) have constant variance across values of independent variables.

4. Case study

4.1. Data description

In this study, the load data of residential individual and commercial building which have totally different load patterns have been used to predict and verify respectively.

The data of residential individual comes from the Residential Energy Consumption Survey Database (RECS) of the Electric Power System which is an open data set. RECS is based on 2010 data on the electricity demand profile of Midwestern American households, which powers the grid and reports to the US government. The data set records the current household appliances and the electricity demand of the entire family every 10 min. The total number of records per household is 52,544. Long-term and large amounts of data allow the model proposed in this study to record and analyze the trends cited in the time series. This study selected five household in the data set for case studies.

The data of commercial building comes from the real case data collected on 2020 by Advanced metering infrastructure (AMI). The frequency of AMI data collection is once every 15 min, and the total number of records is 35,137. Commercial buildings have fluctuations during weekend, and there will also be a significant drop in power consumption during holidays. Electricity consumption is lower in winter and higher in summer.

The CNN-GRU model that we built uses the CNN part to interpret the electricity consumption data map, and then uses the GRU to predict future electricity consumption for data containing time series.

4.2. Model training, validation, testing and prediction interval generating

First part of proposed model (CNN-GRU) utilizes TensorFlowˆ© backend, packages and functions in numpy, pandas, keras, sklearn to help build the proposed model on a computer with Win 10, Python 3.7, Intel Core i5-8400

Table 1. The performance comparison between RELU and SELU.

Activation function	MAE	RMSE	MAPE (%)
RELU	0.0085	0.0221	5.9803
SELU	0.0081	0.0204	3.2814

2.8 GHz CPU, and NVIDIA GeForce RTX 2080 Ti GPU. All codes are programmed in python. The ratio of training, validation and testing are 80%, 15% and 5% of the original data respectively. Also, hyperparameters such as learning rate, epochs, activation function and model structure are also fine-tuned during some experiments.

Through the tuning process, design of experiment had been done to achieve best parameters. Table 1 shows the loss of proposed model using SELU and RELU respectively for residential individuals. Compared with RELU, SELU makes convergence more efficiently, which we can observe from the loss in Table 1. RMSE and MAPE are 0.0005 and 2.3199 lower than using RELU. In this experiment, Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and Mean Absolute Percentage Error (MAPE) are the metrics for the performance evaluation, as shown in (9), (10) and (11). In notation (9) to (11), y represents the true value and $\hat{y}$ for predicted value.

$$MAE = \frac{1}{n}\sum |y - \hat{y}| \tag{9}$$

$$RMSE = \sqrt{\frac{1}{n}\sum (y - \hat{y})^2} \tag{10}$$

$$MAPE = \frac{100\%}{n}\sum \frac{|y - \hat{y}|}{y} \tag{11}$$

In order to generate our predicted quantile, the loss function is changed to quantile regression loss in the CNN-GRU model. Then, according to the predicted results, generate 50% to 90% quantiles, and then use our predictions to obtain 50% to 90% prediction intervals. The proposed model with quantile regression loss function, and finally calculate 5 families. The percentage of true power consumption values in each prediction interval in.

4.3. Results

4.3.1. Point forecasting part- CNN-GRU

We compared the CNN-GRU model with other existing time series forecasting models, such as autoregressive (AR), support vector regression (SVR), LSTM, GRU, and autoregressive integrated moving average (ARIMA). In addition, we combined CNN and LSTM as a hybrid model, and compared with the proposed CNN-GRU model. After simulating the results for 20 times, the highest/lowest and average results are compared with the individuals and commercial building of each model as shown in Table 2 and Fig. 3. Facts have proved that the performance of the model proposed is better than the existing time series point estimation model. Among the five households, the MAE and RMSE of the proposed model are the lowest, and the percentage error of MAPE are also have better performance. For commercial building, the situations are very similar which have better performance for three indicators. In addition, the comparison between the proposed model and the single LSTM and GRU models shows that the CNN part of our proposed model does indeed extract local features of time series data. In addition, the loss between the proposed CNN-GRU and CNN-LSTM models is very close, especially MAE and RMSE. GRU has fewer parameters than LSTM, but it can also implement functions equivalent to LSTM. In terms of practical operation, fewer parameter choices are more beneficial for decision-making, and therefore the results of proposed CNN-GRU will be beneficial to real smart grid operations. Regardless of whether you choose GRU or LSTM, their MAE and RMSE losses are much lower than other existing time series forecasting models.

Artificial intelligence development led to mature deep learning, where CNN is suitable for complicated and high-variety data such as household energy consumption. Combining CNN and GRU as a hybrid model improved time series prediction with better results than other models. Based on the results shown in Fig. 3, it is apparent that the AR model performs poorly in predicting the electricity load of some residential and commercial buildings due to their large variations. While other models, such as LSTM and GRU, can capture the general trend, the proposed CNN-GRU model exhibits better predictive performance when the variability is increased. The CNN-GRU hybrid model outperforms other models, such as LSTM, GRU, ARIMA, SVR, and AR, in electricity load forecasting.

Table 2. Performance comparison between proposed model and existing time series forecasting models.

Sources	Model	MAE (highest/lowest)	RMSE (highest/lowest)	MAPE (highest/lowest) [%]
Household 1 (RECS open data set)	Proposed CNN-GRU	***0.0081(+0.0021/−0.0016)***	***0.0212(+0.008/−0.0094)***	***3.5547(+0.0094/−0.0103)***
	CNN-LSTM	0.0084(+0.0038/−0.0043)	0.0215(+0.011/−0.0086)	3.6950(+0.0094/−0.0103)
	LSTM	0.0085(+0.0041/−0.002)	0.0220(+0.012/−0.0083)	5.9803(+0.5261/−0.6569)
	GRU	0.0084(+0.0044/−0.0028)	0.0217(+0.0125/−0.0122)	3.6604(+0.9804/−0.8066)
	ARIMA	0.0183(+0.003/−0.0033)	0.0311(+0.0148/−0.0136)	47.0051(+1.8079/−1.5167)
	SVR	0.2551(+0.0053/−0.004)	0.3098(+0.0145/−0.0155)	50.5966(+1.9738/−2.0821)
	AR	0.0256(+0.0048/−0.0023)	0.0511(+0.0156/−0.0154)	107.7009(+3.1256/−2.6783)
Household 2 (RECS open data set)	Proposed CNN-GRU	***0.0084(+0.001/−0.0018)***	***0.0239(+0.0087/−0.0094)***	5.6601(+0.6137/−0.6963)
	CNN-LSTM	0.0084(+0.0025/−0.0024)	0.0241(+0.093/−0.01)	5.5215(+0.6992/−0.7129)
	LSTM	0.0093(+0.0037/−0.0049)	0.0243(+0.080/−0.0098)	***5.3056(+0.7976/−0.6036)***
	GRU	0.0086(+0.0047/−0.0052)	0.0261(+0.095/−0.0089)	6.7947(+0.8743/−0.8454)
	ARIMA	0.0198(+0.0045/−0.0057)	0.0440(+0.0146/−0.0130)	47.5592(+1.5564/−1.4147)
	SVR	0.2048(+0.0043/−0.0028)	0.2297(+0.0142/−0.012)	36.8845(+1.6073/−1.9533)
	AR	0.0638(+0.0032/−0.0044)	0.0674(+0.0168/−0.0143)	144.5646(+3.6217/−2.9652)
Household 3 (RECS open data set)	Proposed CNN-GRU	***0.0082(+0.0007/−0.0002)***	***0.0259(+0.0087/−0.0096)***	***1.5071(+0.8904/−0.6266)***
	CNN-LSTM	0.0082(+0.0038/−0.0012)	0.0261(+0.0091/−0.0084)	3.9979(+0.8794/−0.8868)
	LSTM	0.0087(+0.0013/−0.0033)	0.0256(+0.0105/−0.01)	5.5237(+0.9908/−0.8433)
	GRU	0.0083(+0.0049/−0.0029)	0.0259(+0.0113/−0.0101)	3.0358(+0.9222/−0.9804)
	ARIMA	0.0314(+0.0026/−0.0042)	0.0636(+0.0143/−0.0158)	86.6093(+2.2956/−1.6248)
	SVR	0.4063(+0.0053/−0.0057)	0.4419(+0.0150/−0.0156)	78.3690(+2.6816/−3.1074)
	AR	0.0901(+0.0034/−0.0045)	0.1629(+0.0177/−0.0174)	112.1800(+5.1198/−4.3814)
Household 4 (RECS open data set)	Proposed CNN-GRU	***0.0097(+0.0012/−0.0017)***	***0.0239(+0.0089/−0.0082)***	***3.4923(+0.8871/−0.7075)***
	CNN-LSTM	0.0097(+0.002/−0.0011)	0.0240(+0.0116/−0.0112)	4.2357(+0.8367/−0.8124)
	LSTM	0.0101(+0.0038/−0.0033)	0.0246(+0.0115/−0.0107)	8.6479(+0.9971/−0.9471)
	GRU	0.0097(+0.0048/−0.0035)	0.0240(+0.0113/−0.0112)	3.9081(+0.4960/−0.5899)
	ARIMA	0.0252(+0.0051/−0.0042)	0.0485(+0.013/−0.0127)	39.4207(+1.8075/−2.2817)
	SVR	0.0412(+0.0049/−0.0066)	0.0445(+0.0138/−0.0143)	82.5762(+3.0655/−5.5985)
	AR	0.1819(+0.0042/−0.005)	0.1872(+0.0162/−0.0153)	190.5713(+7.8222/−5.4965)
Household 5 (RECS open data set)	Proposed CNN-GRU	0.0233(+0.001/−0.0024)	0.0074(+0.0018/−0.0014)	***2.1297(+0.5206/−0.5366)***
	CNN-LSTM	0.0236(+0.003/−0.0031)	0.0075(+0.0017/−0.0015)	2.3834(+0.8652/−0.5275)
	LSTM	0.0245(+0.0031/−0.0033)	0.0078(+0.0012/−0.0018)	3.6766(+0.6223/−0.5680)
	GRU	0.0241(+0.0042/−0.0048)	***0.0074(+0.0015/−0.0012)***	2.2478(+0.6799/−0.6030)
	ARIMA	***0.0227(+0.0044/−0.0064)***	0.0549(+0.0149/−0.0127)	31.2541(+1.8725/−1.5351)
	SVR	0.3144(+0.005/−0.0051)	0.3482(+0.0129/−0.0136)	25.4797(+1.4001/−1.8434)
	AR	0.0260(+0.0053/−0.0038)	0.0451(+0.0141/−0.016)	129.1985(+5.8766/−6.0005)
Commercial Building (AMI data)	Proposed CNN-GRU	9.6457(+2.5007/−1.9053)	***13.4981(+5.0936/−5.9850)***	***1.1738(+0.0031/−0.0034)***
	CNN-LSTM	***9.5639(+4.3265/−4.8958)***	13.9552(+7.1398/−5.5820)	1.3287(+0.0033/−0.0037)
	LSTM	10.3979(+5.0154/−2.4465)	15.0233(+8.1945/−5.6678)	1.4193(+0.1248/−0.1559)
	GRU	11.4477(+5.9964/−3.8159)	15.9427(+9.1835/−8.9631)	1.6713(+0.4476/−0.3667)
	ARIMA	52.2630(+8.5677/−9.4244)	66.1065(+31.4590/−28.9083)	12.3218(+0.8439/−0.7513)
	SVR	53.8035(+9.7641/−8.1344)	62.8835(+30.6415/−29.4615)	13.1814(+1.1281/−1.6421)
	AR	379.3941(+71.1363/−34.0861)	395.3860(+120.7049/−119.1574)	88.6577(+3.6421/−4.2153)

Unlike LSTM and GRU, the proposed CNN-GRU model can capture both spatial and temporal features, which is particularly useful in predicting the electricity load of varying behaviors. While the ARIMA model is appropriate for predicting stationary time series data based on statistical analysis, it is not suitable for forecasting the non-linear and dynamic behavior of electricity load. The SVR model, which is a machine learning model that can handle non-linear data, is not optimal for time series data with complex temporal patterns. The AR model, which is a simple autoregressive model that only considers past values of the target variable, cannot capture the complex dependencies and interactions between electricity demand and other factors. Overall, the CNN-GRU model's ability to capture both spatial and temporal features, along with its flexibility in handling non-linear and dynamic behavior, make it a promising model for electricity demand prediction.

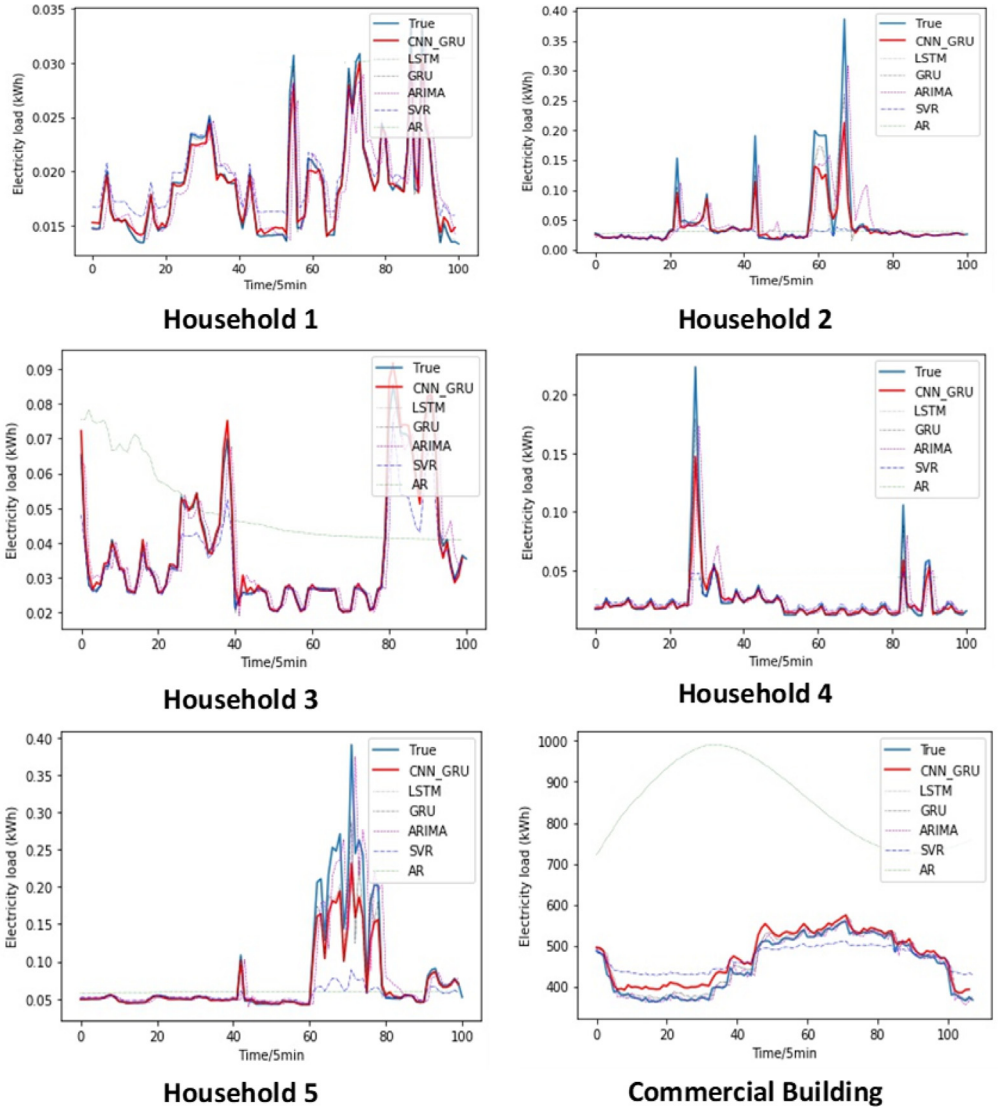


Fig. 3. Comparison with each model for average value of household 1 to 5 and commercial building (from upper left to lower right).

4.3.2. Prediction interval part—50% to 90% interval using quantile regression loss function

With the result from point forecasting, 5%, 10% 15%, 20%, 25%, 75%, 80%, 85%, 90% and 95% quantiles are generated to produce the 50% to 90% prediction interval using our proposed model with quantile regression loss function. The coverage of 50% to 90% prediction interval for each household and commercial building are shown in Table 3. The rate of covering true consumption value shows the prediction interval that correctly covers the true power consumption value. Take the 50% prediction interval in household 1 as example, each predicted value will be given a 50% prediction interval generated by quantile regression loss function, and 95.04% of the 50% prediction interval contains the true power consumption value. The difference in Table 3 refers to the difference between the true consumer value coverage rate and the predicted coverage rate. The reliable point forecasts are difficult to produce true power consumption of residential household, but probabilistic load forecasting has the ability to catch the volatility and uncertainty by intervals, density, or quantiles. From the results in Table 3, we can see that all the results of covering true consumption value in each prediction interval of different household contains broader than expected.

Table 3. Performance evaluation of prediction interval for each household and commercial building.

Sources	Prediction interval	Rate of covering true consumption value	Difference %	Sources	Prediction interval	Rate of covering true consumption value	Difference %
Household 1 (RECS open data set)	50%	95.04%	45.04	Household 4 (RECS open data set)	50%	83.48%	33.48
	60%	97.92%	37.92		60%	97.65%	37.65
	70%	99.10%	29.10		70%	99.91%	29.91
	80%	99.91%	19.91		80%	99.37%	19.37
	90%	99.82%	9.82		90%	99.91%	9.91
Household 2 (RECS open data set)	50%	85.56%	35.56	Household 5 (RECS open data set)	50%	95.49%	45.49
	60%	96.66%	36.66		60%	95.49%	35.49
	70%	97.29%	27.29		70%	98.19%	28.19
	80%	99.01%	19.01		80%	98.92%	18.92
	90%	99.82%	9.82		90%	97.38%	7.38
Household 3 (RECS open data set)	50%	94.22%	44.22	Commercial Building (AMI data)	50%	93.72%	43.72
	60%	97.20%	37.20		60%	94.41%	34.41
	70%	98.74%	28.74		70%	98.13%	28.13
	80%	99.28%	19.28		80%	99.46%	19.46
	90%	99.82%	9.82		90%	99.35%	9.35

In Table 3, we can observe that either 50% or 90% prediction interval contains more true power consumption value than expected. In smart grid, this kind of information is not only useful for power provider but also power consumer. It helps decision makers of power providers to determine the policies based on a prediction interval instead of single predicted value but also RE100 company to realize the risk of renewable energy purchasing plan. It also helps power consumers distribute their power in the future of smart grid era through the disclosure about the uncertainty of the predicted power consumption, and reduce the waste of power, the unequal supply and demand of power. This technology is significantly beneficial for improving the ability of estimating the power consumption in smart cites.

Probabilistic forecasts are valuable in revealing the uncertainty in predictions and aiding decision makers in making optimal choices. Previous literature has demonstrated the effectiveness of probabilistic forecasting in addressing energy-related issues that involve randomness and intermittency. In this study, deep learning and probabilistic forecasting were combined to create prediction intervals for household and commercial buildings in a smart grid. By using the CNN-GRU model, hidden patterns in time series data were identified, resulting in more accurate predictions compared to existing time series forecasting models. Additionally, probabilistic forecasting combined with deep learning demonstrated excellent performance. The prediction intervals generated by the CNN-GRU model contained broader and more accurate power consumption values than expected, providing uncertainty information to both power consumers and suppliers in the smart grid. The hybrid model has been verified to be more effective than standalone models.

5. Conclusions

Short-term forecasting of individual household and commercial buildings is an important and challenging topic. In a smart grid, this type of information is not only useful to electricity providers, but also to electricity consumers. It can help power supplier decision makers to determine strategies based on forecast intervals rather than individual forecast values but also RE100 company to realize the risk of renewable power purchasing. It also discloses the uncertainty about predicted power consumption to help power consumers allocate power in the future of the smart grid era and reduce power waste and uneven power supply and demand.

Furthermore, deep learning has not explored extensively is the uncertainty. Therefore, occasionally something beyond a point estimate is required to decide. This study proposed a novel probabilistic hybrid deep learning neural network framework combining CNN and GRU with quantile regression loss to deal with univariate and volatile building power consumption forecasting. From the perspective of point estimation, although high performance can be achieved by LSTM or GRU network alone, we furtherly demonstrate that the hybrid framework that was proposed in this study outperforms the conventional LSTM and GRU by the numerical study from real case. The CNN extracts

the most useful information from the original raw data, and GRU store the information in time series. From the perspective of interval estimation, due to the univariate and uncertainty of building power consumption, we choose quantile regression loss function to help generate prediction interval. When estimating the electricity consumption, the consumption of an individual household and commercial building are varying a lot. But with the proposed method, power providers are able to decide their electricity production under different type of users and adjust their electricity (such as sell or buy to/ from others). In other words, probabilistic intervals can provide customized and differentiated decision-making for grid dispatching. Electricity will not be wasted and power demand control can be achieved.

In summary, the hidden patterns in time series data could be extracted by combination of CNN and GRU, and the forecast could be more accurate than other existing time series forecasting models. Also, implementing probabilistic with deep learning did have great performance. Generated prediction intervals based on CNN-GRU can contain broader, and more true power consumption value are contained in the prediction interval than expected which enhanced credibility. As a result, the proposed method can contribute to uncovering the uncertainty in the prediction and thus helping the decision makers take optimal decisions. In previous study in probabilistic forecasting, abundant literature prove probabilistic forecasting can tackle randomness and intermittence involved energy related issues. In this study, probabilistic forecasting is combined with deep learning method, the accurate results of deep learning method make probabilistic forecasting result perform better than expected. The limitation is that in order to get different quantiles, CNN-GRU model must be run for several times, which is time consuming.

The limitation of the research is that the amount of data should be recorded for a longer period, so that the hidden electricity consumption trends can be analyzed. Some improvements can be made in the future. First, more data from different power consumption features should be added to multivariate prediction. For example, adding data from temperature, humidity, weather, holidays, rental status, etc. Second, we can try to increase the amount of data. Currently, the data is collected for one year. If We want to add seasonal characteristics, we may need the data at least 3 to 5 years. Third, the CNN-GRU can be trained, tested and verified on real-time data to meet the needs of smart grid monitoring. Furthermore, other hyperparameter adjustment methods can also be used for the proposed model.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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